

EARTH



WE LIVE ON A GIANT BALL OF ROCK spinning round the Sun, which we call the Earth. The Earth is one of nine planets in the Solar System and one of the four made of rock.

However, the Earth is unique, because it is the only planet in the Solar System – and perhaps even in the Universe – that can support life. The distance of the Earth from the Sun makes it neither very hot like Venus, nor icy cold like Pluto, enabling liquid water to exist on its surface. The Earth also has an oxygen-rich atmosphere. These two substances – water and oxygen – are the key factors that allow life to flourish on the Earth.

Structure of the Earth

By recording the way vibrations from earthquakes reverberate through the Earth, scientists have discovered that the Earth has an egg-like structure. At its centre is a “yolk” of metal, surrounded by an “egg-white” of soft rock called the mantle, and an outer “shell” of hard rock called the crust.



The Earth's crust consists of a number of interlocking slabs of rock called tectonic plates.

Solid iron

Earth's ingredients

Although more than 80 elements (basic substances) occur naturally on the Earth, the bulk of the Earth is made of iron (35%), oxygen (28%), magnesium (17%), and silicon (13%). The following elements are present in significant, but small, amounts: nickel (2.7%), sulphur (2.7%), calcium (0.6%), and aluminium (0.6%). Tiny proportions of other elements make up the remainder (0.6%).

Molecule of oxygen gas

Magnesium ore (magnesite)

Locket containing crystal of silicon

Nickel ore (nickeline)

Sulphur crystals

Meteorites

Meteorites are natural objects that fall to the Earth from space. They are made of iron, stone, or a mixture of both. The two main types of meteorite are called chondrites and achondrites.

Calcium-rich chalk

Aluminium ore (bauxite)

Chondrite meteorite

Achondrite meteorite

The Earth's structure

Atmosphere is a thin surrounding layer of gases about 640 km (400 miles) deep.

Crust, Earth's outer layer of rock varies in thickness: beneath the oceans, it is 6–11 km (4–7 miles) thick, but it stretches up to 70 km (43 miles) under mountain ranges.

Mohorovicic discontinuity, or **Moho**, is the boundary between the crust and the mantle.

Mantle is a partially molten layer beneath the crust, extending to a depth of about 2,900 km (1,800 miles) and made largely of a rock called peridotite.

Gutenberg discontinuity is the boundary between the mantle and the core.

Outer core reaches to a depth of about 4,900 km (3,050 miles) and is made of molten iron and nickel – magnetic metals that give the Earth its magnetic field.

Inner core, like the outer core, is made of iron and nickel, but although temperatures reach 3,700°C (6,690°F), the pressure is so great that the metal remains solid.

Richard Oldham

By examining the seismographic recordings of earthquakes, the British geologist Richard Oldham (1858–1936) discovered that earthquakes produce two different kinds of vibration. He called them primary (P) waves and secondary (S) waves. Oldham's analysis revealed that P waves travel more slowly through the core of the Earth than through the mantle. He concluded that Earth's core must be liquid, which is partly true.

